

Yuankun Huang

Software Engineer — Game Clients & Real-Time Systems

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Client & systems engineer with **5+ years** shipping a live-service title from **early development to global launch** (10M+ downloads). I build the parts of games where timing, determinism, and memory budgets are non-negotiable — proven by **CI-enforced engineering contracts**, not claims.

CORE SKILLS

Languages	C# (expert), C++20, Python
Engines & Runtime Systems	Unity (production, shipped), .NET 8, SFML, miniaudio Client networking & state sync (Protobuf, SignalR, MessagePack), deterministic record/replay, audio-clock timing pipelines, hot-update (ILRuntime, Addressables)
Performance	Zero-allocation hot paths, object pooling, struct-oriented ECS, GPU instancing, Unity Profiler / Frame Debugger, low-end device optimization
Build & Quality Certification	CMake, vcpkg, doctest, GitHub Actions CI, headless test gates, custom allocation guards Unity Certified Expert: Programmer

PROFESSIONAL EXPERIENCE

Unity Developer May 2025 – Present
IGG Canada — Live Service & New-Title Client Development Vancouver, BC

- Rejoined **Mythic Heroes** live ops and delivered a wave of 2025 live-event and monetization systems — a full **pet progression suite** plus flip-card, blind-box, event-calendar, and event-pass (**25 UI windows, 270+ C# files**) via monthly **ILRuntime** hot-updates; built **Analyze+**, an Addressables duplicate-reference analyzer.
- On an unannounced strategy title, architecting the **live-event client framework**: data registries, red-dot tree, calendar scheduling, and a content factory powering **20+ event types** with editor prefab generators; primary owner of two game systems.

Career Break: Bereavement & Family Care

Feb 2025 – May 2025

Software Engineer (Contract, Hybrid)

Jan 2024 – Jan 2025

The HIVE, UBC Faculty of Medicine — Gamified Rehabilitation Platform

Vancouver, BC

- Sole engineer** for TrAIT, a clinically validated Unity rehabilitation system (Leap Motion hand-tracking) with adaptive difficulty (Challenge Point Framework) — owned all engineering, requirements through deployment.
- System achieved **~1,000 movements/session** vs. 32 in traditional therapy, supporting published results showing **~13% functional improvement** (*Neurorehabilitation and Neural Repair*).

Unity Developer

Sep 2020 – Sep 2024

IGG Canada — Mythic Heroes (10M+ downloads)

Vancouver, BC

- Core client engineer from early development to **global launch**; implemented client-side networking and multiplayer state synchronization (**Google Protobuf**) for live features — primary client owner of **Chat**, top-2 contributor on **Guilds**; shipped monthly content via **ILRuntime** hot-updates throughout live ops.
- Stood up the client network module for **Dawn of Survival**, a second shipped title: WebSocket socket/message managers, **Protobuf** serialization, gzip HTTP session layer, and pluggable login adapters.
- Co-developed the shared UI framework spanning **~600 screens** game-wide, personally authoring **150+** (navigation, screen flow).
- Profiling-driven optimization for low-end devices (Unity Profiler, Frame Debugger; AssetBundle restructuring) — part of a team push that cut load times **~30%**, build size **~20%**, and held stable **60 FPS** on mid-range Android for launch.

SELECTED PROJECTS

Rhythm Fruit Shop — Native C++ Game Core [GitHub]

C++20 / SFML / miniaudio / CMake / CI

- Built a real-time rhythm core where song time derives from the **audio backend's PCM cursor** through an **EMA-smoothed clock** (not frame deltas); input events carry host-monotonic timestamps reverse-mapped to song time for judgment-grade accuracy.
- Implemented **deterministic record & replay** on a headless session core, validated in CI against an **independent Q16 fixed-point oracle**; **zero-allocation hot path** (custom operator `new AllocationGuard`) and layering enforced via **CMake dependency guards**.

Kodama — Server-Authoritative Multi-Agent Simulation [GitHub]

.NET 8 / SignalR / MessagePack / Unity

- Built a distributed real-time simulation sustaining **10K agents** with sub-millisecond server ticks: struct-oriented ECS, spatial partitioning, zero-GC hot path, and a dumb-client renderer using GPU instancing; binary sync cut payloads **~70%** vs. JSON.

Synapse — Cross-Language Distributed Simulation [GitHub]

.NET 8 / Python / Unity

- Streamed **2,000+ Python-driven agents** into a Unity client at stable **60 FPS**, using lock-free concurrency, pooling, and state conflation to hold per-update managed allocations near **1.4 KB**.

EDUCATION

M.S. Computer Science (Computing Systems), Georgia Institute of Technology (OMSCS, part-time online) Starting Fall 2026

Master of Digital Media, Centre for Digital Media, Vancouver

2021

B.Eng. Telecommunication Engineering, BUPT / Queen Mary University of London

2019